Question:

John buys milk for 2 dollars, eggs for 3 dollars, light bulbs for 3 dollars, cups for 3 dollars, and roach traps for 4 dollars. If there is a 10% tax on all nonfood items how much did everything cost.

Reasoning + Answer:

<think> To solve this problem, let's first identify the items that have a tax applied to them and then calculate the total cost including the tax. From the information provided, we can see that there are two types of items: food items (milk, eggs, cups) and nonfood items (light bulbs, roach traps). Let's separate them first. Now, let's focus on the nonfood items: light bulbs and roach traps. According to the problem, there is a 10% tax on these items. The cost of light bulbs is 3 dollars and the cost of roach traps is 4 dollars. To calculate the tax for each nonfood item, we can use the following formula: \(\text{Tax} = \text{Item Cost} \times \text{Tax Rate} \). For light bulbs, the tax would be \(3 \times 0.10 = 0.3 \) dollars. For roach traps, the tax would be \(4 \times 0.10 = 0.4 \) dollars. Next, we need to add the tax to the original cost of each nonfood item to find the total cost for each of these items including tax: - For light bulbs: \(3 + 0.3 = 3.3 \) dollars. - For roach traps: \(4 + 0.4 = 4.4 \) dollars. Now, let's sum up the costs of all items including the tax. We also need to add the costs of the food items (milk and eggs) since they are not taxed. The costs of food items are: milk (2 dollars) and eggs (3 dollars). So, let's add the costs of nonfood items with tax and food items: - Total cost for light bulbs and roach traps including tax: \(3.3 + 4.4 = 7.7 \) dollars. - Total cost for food items: \(2 + 3 = 5 \) dollars. Finally, to find the overall total cost, we add the total cost of nonfood items including tax and the total cost of food items together: - Overall total cost: \(7.7 + 5 = 12.7 \) dollars. </think><answer> 12.7 </answer>